Assignment 1 - Git & GitHub Fundamentals

Session 1 - Group D - Individual Lab Work

REPORT

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Course	Software Engineering
Course Code & Section	CSC 3324 - Section 03
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1. Repository Setup

I created the project git-session1-zfenjiro and initialized it as a Git repository. In .gitignore, I ignored local.env and normal .log files while keeping important.log visible to Git with an exception rule. I used Git status and ignore checks during the setup to verify which files were tracked or ignored.

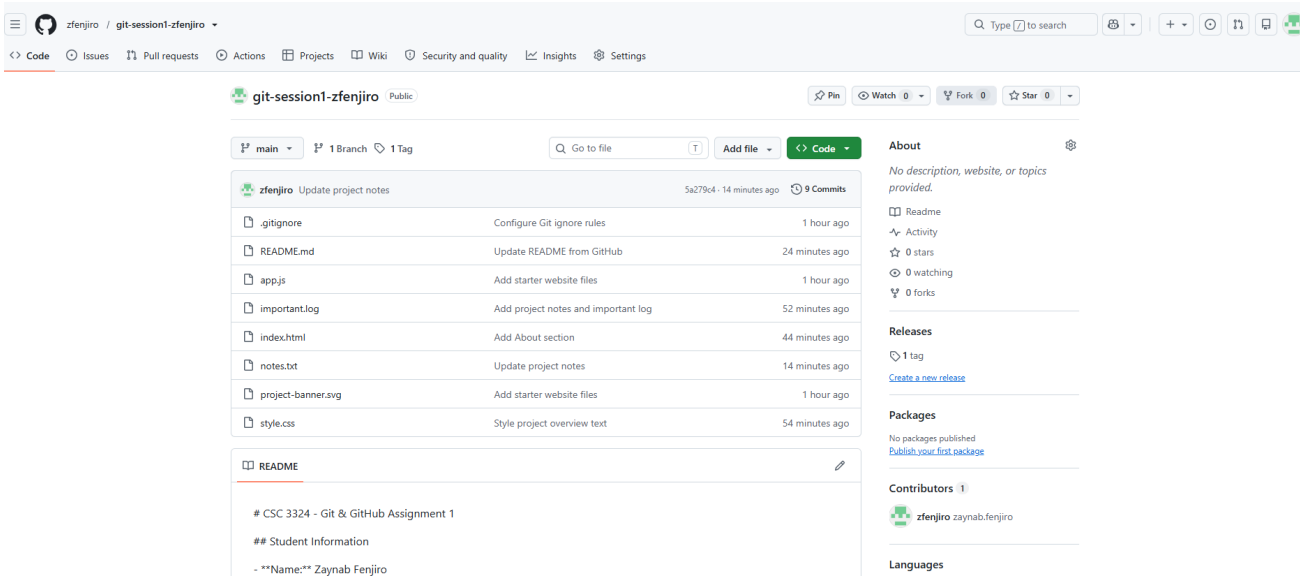


Figure 1. Final public GitHub repository showing the project files, main branch, commit count, branch, and tag.

2. README & Commit History

I created README.md with my student information, a short project description, the technologies used, an example Git command, a link to my GitHub profile, the provided project banner reference, and a Remote Update section added during the remote-change exercise. I organized the work into multiple logical commits with meaningful messages rather than putting all changes into one commit.

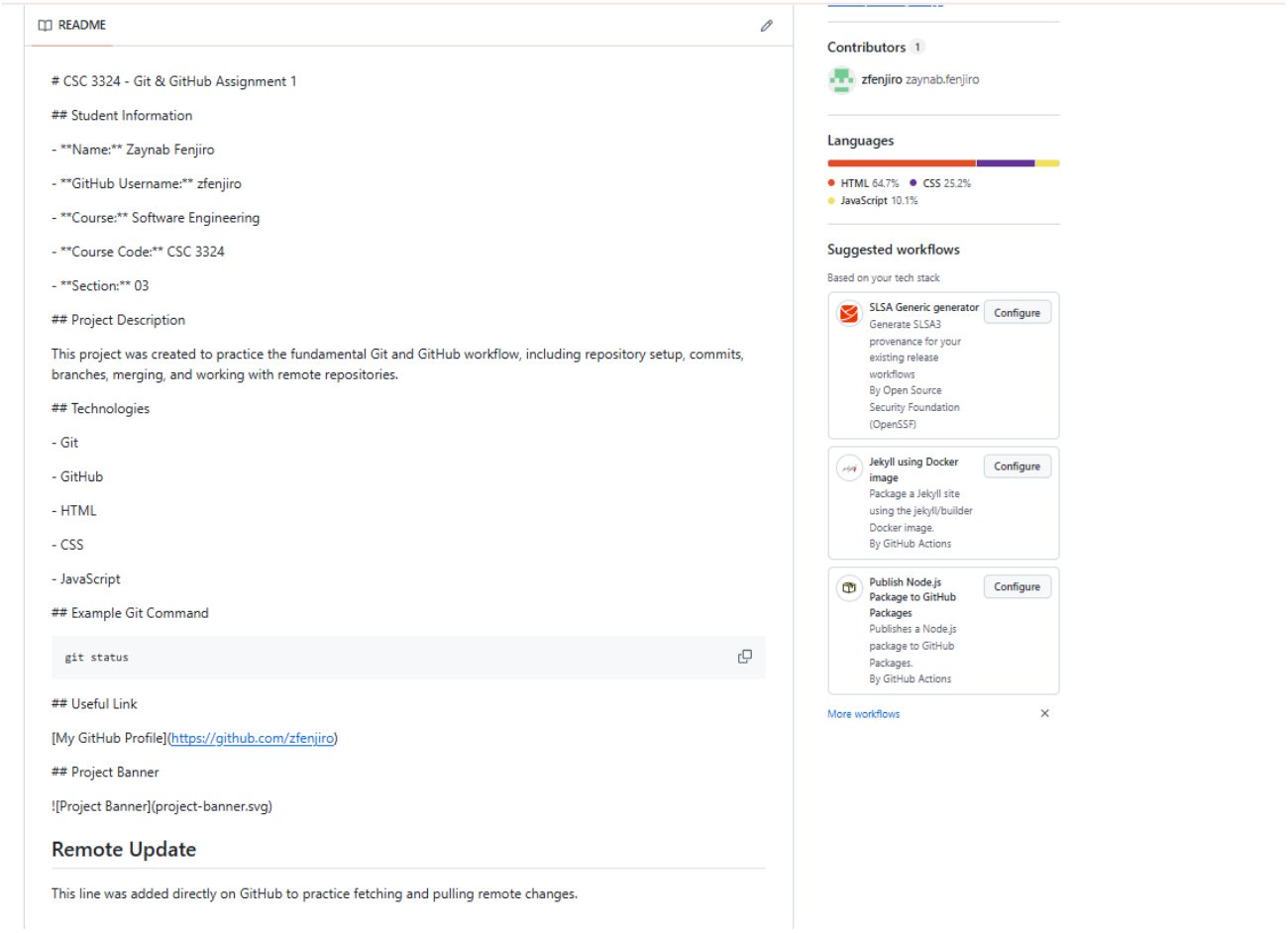


Figure 2. README displayed on the GitHub repository page.

```
C:\Users\Administrator\Documents\git-session1-zfenjiro>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\Administrator\Documents\git-session1-zfenjiro>git log --oneline --all --graph --decorate
* 5a279c4 (HEAD -> main, origin/main, origin/HEAD) Update project notes
* 89e5a67 (tag: v1.0) Update README from GitHub
* 5f3c27a (about-feature) Add About section
* 3047a15 Add project notes and important log
* 267760e Style project overview text
* 1f65449 Update project overview
* 5e2361d Add starter website files
* c0b8f08 Add project README
* fcc712d Configure Git ignore rules

C:\Users\Administrator\Documents\git-session1-zfenjiro>
```

Figure 3. Compact Git history showing nine commits, the main branch, about-feature, origin/main, and tag v1.0.

3. Selective Staging

I modified index.html and style.css at the same time. I then staged only index.html with git add index.html. Git status showed index.html under “Changes to be committed” while style.css remained under “Changes not staged for commit.” I committed index.html first and later staged and committed style.css separately. This demonstrated how Git can separate related working-directory changes into different logical commits.

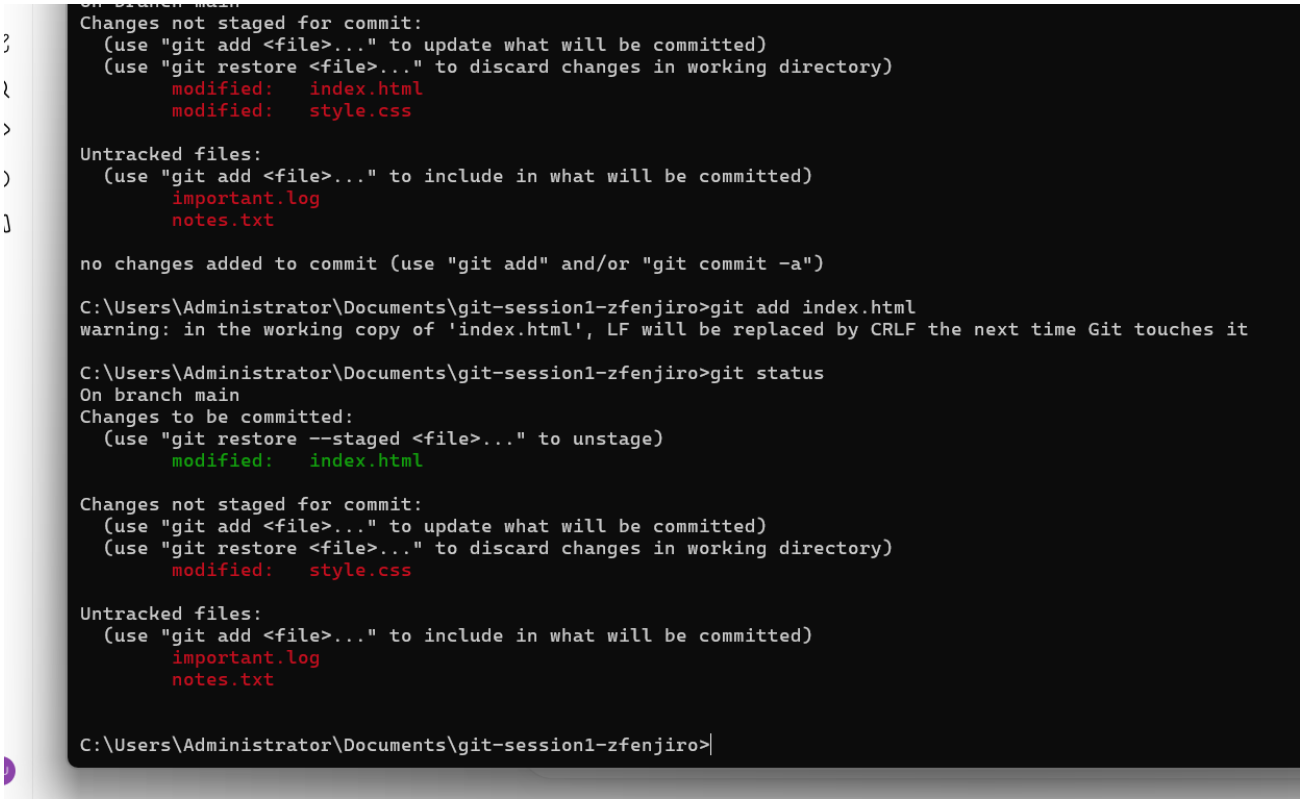


Figure 4. Selective staging: index.html is staged while style.css remains modified and unstaged.

4. Branch & Merge

I created a feature branch named about-feature and used it to add an About section. I committed the feature on that branch, returned to main, and merged the feature into main. The history confirms the Add About section commit and shows the about-feature branch reference.

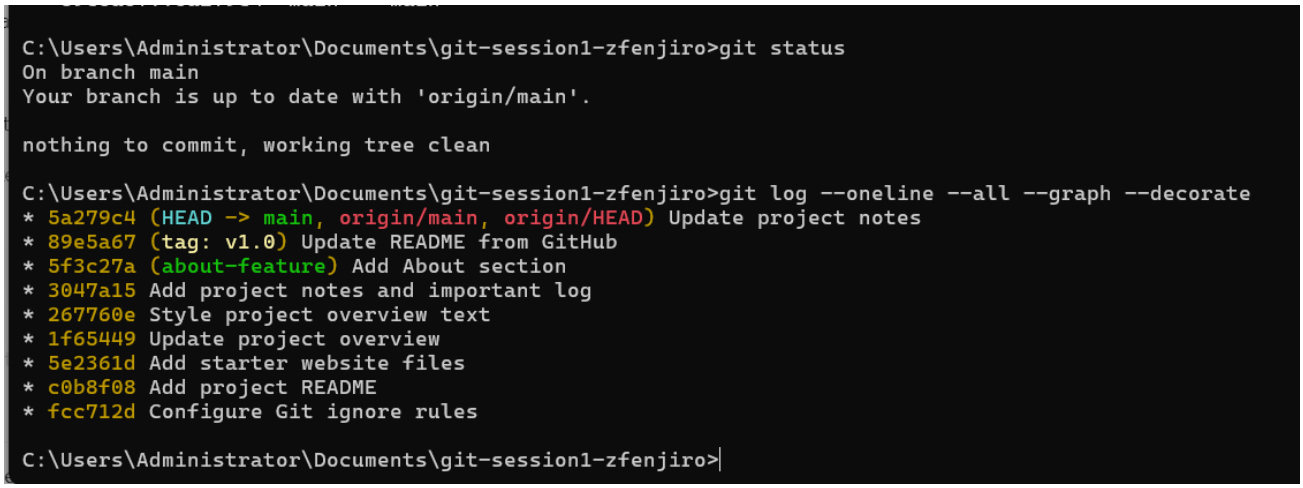


Figure 5. Git history confirming the about-feature branch and the Add About section commit.

5. Remote Changes

A commit records a snapshot of staged changes in the local Git repository, while push sends local commits to the remote GitHub repository. Fetch downloads remote updates without changing the current local working files or automatically integrating them into the current branch. Pull retrieves remote changes and integrates them into the local branch. I edited README.md directly on GitHub, ran git fetch origin, and used git status to verify that local main was behind origin/main by one commit. I then ran git pull origin main,

which fast-forwarded the local branch and updated README.md. I also practiced unstaged and staged diff commands during the assignment.

```
C:\Users\Administrator\Documents\git-session1-zfenjiro>git push -u origin main
Enumerating objects: 25, done.
Counting objects: 100% (25/25), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (25/25), 3.36 KiB | 313.00 KiB/s, done.
Total 25 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), done.
To https://github.com/zfenjiro/git-session1-zfenjiro.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\Administrator\Documents\git-session1-zfenjiro>git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.01 KiB | 35.00 KiB/s, done.
From https://github.com/zfenjiro/git-session1-zfenjiro
   5f3c27a..89e5a67  main       -> origin/main

C:\Users\Administrator\Documents\git-session1-zfenjiro>git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean

C:\Users\Administrator\Documents\git-session1-zfenjiro>git pull origin main
From https://github.com/zfenjiro/git-session1-zfenjiro
 * branch            main       -> FETCH_HEAD
Updating 5f3c27a..89e5a67
Fast-forward
 README.md | 3 +++
 1 file changed, 3 insertions(+)

C:\Users\Administrator\Documents\git-session1-zfenjiro>git status
```

Figure 6. Remote workflow showing push, fetch, local main behind origin/main by one commit, and the subsequent pull.

6. Fair Use of AI

I used ChatGPT occasionally when I needed clarification about a few Git commands and to check that I was following the assignment steps correctly. For example, I asked for help understanding the difference between fetch and pull and how to verify staging and branch status. I ran the commands myself, reviewed the terminal output, and checked the results on GitHub before continuing. I only used the explanations as guidance and made sure I understood the commands included in my submission.

7. Sources & Reflection

Sources used: the CSC 3324 Assignment 1 instructions and report template; Git and GitHub output from my own repository; GitHub; and ChatGPT for explanations and step-by-step guidance.

The hardest concept for me was understanding the difference between the local repository and the remote repository, especially fetch versus pull. I learned that fetch lets me inspect remote changes before integrating them, while pull brings those changes into the local branch. If Git behaves unexpectedly, I would first run git status and then check the current branch, recent commit history, and remote state before making another change.